

MATH 2R03 - Assignment 2

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Question 1

Let V be a vector space over a field $\mathbb{F}$, and let $v_1, \dots, v_m$ be a list of vectors in V . For $k = 1, \dots, m$ set $w_k := v_1 + \dots + v_k$. Show that $\text{span}(w_1, \dots, w_m) = \text{span}(v_1, \dots, v_m)$.

Solution:

Question 2

Prove or give a counterexample: $v_1, \dots, v_m$ is a linearly independent list of vectors in V , then $5v_1 - 4v_2, v_2, \dots, v_m$ is linearly independent.

Solution:

Question 3

Let $v_1, \dots, v_m$ be a linearly independent list of vectors in V , and let $w \in V$. Show that if the list $v_1, \dots, v_m, w$ is a linearly independent list if and only if $w \notin \text{span}(v_1, \dots, v_m)$.

Solution:

Question 4

Let $\mathbb{F}$ be a field. For $n \in \mathbb{N}$, let:

$$\mathbb{F}[x]_n := \{ p \in \mathbb{F}[x] : \deg(p) \leq n \}$$

(a) Prove that $\mathbb{F}[x]_n$ is a subspace of $\mathbb{F}[x]$.

Solution:

(b) Explain why there does not exist a list of six polynomials that is linearly independent in $\mathbb{F}[x]_4$

Solution:

(c) Explain why no list of four polynomials spans $\mathbb{F}[x]_4$.

Solution:

Question 5

Consider the set $F_3 := \{0, 1, 2\}$ together with the addition $a \oplus b := (a + b) \bmod 3$ and the multiplication $a \otimes b := (ab) \bmod 3$.

(a) Verify that $\oplus$ and $\otimes$ satisfy the field axioms; only submit here the proof of distributivity.

Solution:

(b) List all functions in the vector space $F_3^{F_3}$ (how many are there?), and determine which of these are polynomial functions.

Solution:

(c) Give a linearly independent list of polynomial functions that spans $\mathcal{P}(F_3)$.

Solution: